

Operating Systems Assignment 1

Instructions

1. Attempt all the questions
2. Type your Name (starting with Sir name as registered) and your Admission Number in a Word document
3. Submit on the Assignment 1 Submission Link

Question 1

Using Strathmore university as a case study, discuss how operating systems manage multi-user environments in such an educational setting, Focus on user management, resource allocation, and security. Give relevant examples in this context. (6 marks)

Question 2

The following processes are being scheduled using a preemptive, roundrobin scheduling algorithm.

<u>Process</u>	<u>Priority</u>	<u>Burst</u>	<u>Arrival</u>
P_1	40	20	0
P_2	30	25	25
P_3	30	25	30
P_4	35	15	60
P_5	5	10	100
P_6	10	10	105

Each process is assigned a numerical priority, with a higher number indicating a higher relative priority. In addition to the processes listed below, the system also has an **idle task** (which consumes no CPU resources and is identified as *Pidle*). This task has priority 0 and is scheduled whenever the system has no other available processes to run. The length of a time quantum is 10 units. If a process is preempted by a higher-priority process, the preempted process is placed at the end of the queue.

- a. Show the scheduling order of the processes using a Gantt chart. [3 marks]
- b. What is the turnaround time for each process? [4 marks]
- c. What is the waiting time for each process? [4 marks]
- d. What is the CPU utilization rate? [2 marks]